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(PROBABLY ALMOST)

EVERYTHING ON THE SYLLABUS

XBI11 IAL EDEXCEL BIOLOGY

UNIT 1 & 2

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Proteins and Enzymes

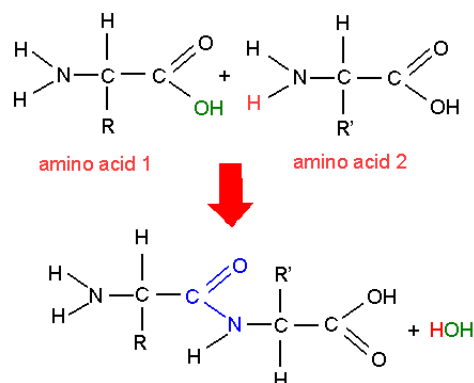
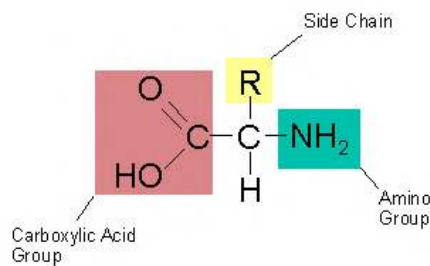
Proteins 🧬

▼ What are proteins?

Complex macromolecule polymers of amino acids

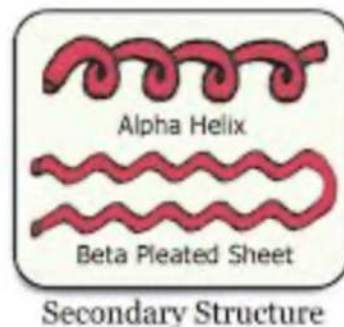
▼ Describe how proteins are formed

- Formed by the **polymerisation of amino acids** in a series of **condensation reactions** to form a **peptide bond** between the carboxyl group of one amino acid and the amino group of another, with **the removal of water**.
- Amino Acid Structure:



▼ Differentiate between the 4 different structures of proteins

- The structure of proteins is determined by the **order and number** of amino acids, the **bonding present** and the **shape** of the protein:
1. **Primary structure:** **The sequence of amino acids** joined by peptide bonds **in the polypeptide chain**.
 2. **Secondary structure:** The primary polypeptide chain coiling into an alpha helix or folded into a beta pleated sheet by the formation of hydrogen bonds.



- The **tertiary structure** is formed when more bonds such as hydrogen, ionic and disulphide bonds are formed between amino acid molecules, causing them to coil or fold further to form fibrous or globular proteins.
- Some proteins have a **quaternary structure**, this is when **2 or more polypeptide chains** are joined together, sometimes with the addition of a **non-protein prosthetic group**. Is determined by the bonds in the tertiary structure.

▼ Differentiate between **Fibrous** and **Globular** Proteins (Structure and functions)

Globular Proteins

- **Round, Compact, soluble** 3D proteins made up of **multiple polypeptide chains** coiled together
- The hydrophilic part faces the outside making the protein **soluble**, and thus easily transported in fluids.
- Haemoglobin is a **water soluble globular protein** which consists of **two beta polypeptide chains, 2 alpha polypeptide chains and 4 haem groups**. It carries oxygen in the blood as oxygen can bind to the haem (Fe^{2+}) group, and oxygen is then released when required. Haemoglobin

is made of 4 polypeptide chains as well as 4 prosthetic groups, so it has a quaternary structure.

Fibrous Proteins

- **Mainly secondary structure** formed by long polypeptide chains **crosslinked** between each other in a repeated amino acid sequence that makes it **strong**
- Collagen is a fibrous protein of great strength due to presence of both hydrogen and covalent bonds in its structure. Collagen molecules wrap around each other and form fibrils which form strong collagen fibres. Collagen forms the structure of bones, cartilage and connective tissue and is a main component of tendons which connect muscles to bones

▼ What are the 4 levels of protein structure? Explain each

1. Primary - Sequences of Amino Acids
2. Secondary - Can be twisted into alpha helixes or Beta pleated sheets. This is due to the attraction of the O from C=O and H from NH. This is stabilized by hydrogen bonds.
3. Tertiary - The secondary structure is folded into the tertiary 3D structure. This is due to the attraction by the side R groups.

Bonds such as hydrogen bonds form due to hydrophilic or polar groups - these can be broken by high temperatures.

There is also hydrophobic interaction by non polar R groups.

Ionic Bonds can form between R groups of amine & carboxyl and these can be broken by pH changes.

Disulphide bridges can only form between cysteine molecules (have S) and can be broken by reducing agents such as tris (2-carboxyethyl) phosphine (TCEP).

4. Quaternary Structure - 2 or more polypeptide chains combine due to the attraction of R groups. Can mix between alpha chains and beta pleated sheets. Can have prosthetic groups in Globular Protein.

▼ What are the 2 types of protein? State their characteristics.

Fibrous

- Long, parallel chains

- Crosslinks between molecules form strong fibers
- Amino acids have non polar, hydrophobic groups so cannot be soluble in water.
- Stable
- Used for mechanical & structural support - keratin, collagen

Globular

- Chains are folded into a compact, spherical shape
- Hydrophobic groups face inwards & hydrophillics are outwards
- Soluble & Unstable in structure
- Examples - Enzymes, Insulin, Hormones & antibodies
- Can be conjugated proteins - have a prosthetic group (non-protein) attached - Haemoglobin

Enzymes 🍌

▼ Describe the structure and function of enzymes

- Enzymes are biological catalysts that increase the rate of reaction by lowering the activation energy (the energy needed for a reaction to occur) of the reactions they catalyse.
- **Active site** - region where the substrate molecules bind to the enzyme to undergo the reaction. Since enzymes are proteins, they have a very **specific 3D shape** due to the bonding in their tertiary structure. This means enzymes have a **specific and complementary shape** to the substrates they bind to, meaning that only **one type** of substrate fits into the active site of the enzyme.
- When the enzyme and substrate form a complex (an **enzyme-substrate complex**), the tertiary structure of the enzyme is **altered** so that the active site of the enzyme fits around the substrate. This is called the **induced fit model**. Once an enzyme-substrate complex is formed and there is **sufficient energy**, the reaction can take place, often breaking down the substrate or combining 2 to build up larger molecules.

▼ Exam Questions

- Describe how the human lung is adapted for rapid gas exchange.

Ans: The human lung has a large number of alveoli which provide a large surface area for gas exchange to occur through diffusion. The walls of the alveoli are thin and are made up of a single layer of flattened cells that ensures a short diffusion distance. There is also an extensive network of capillaries that maintain the steep concentration gradient for gas exchange to occur.

Question Number	Answer	Additional Guidance	Mark
1(b)	(QWC – Spelling of technical terms must be correct and the answer must be organised in a logical sequence) (folded into) large number of alveoli ; (providing) large surface area ; - walls of {alveoli / capillary} are thin ; (the walls) are made from a single layer of flattened cells ; - idea that (thin walls) ensures short diffusion distance ; - idea of a (extensive) capillary network ; - maintains a concentration gradient ;	(QWC emphasis on clarity of expression) IGNORE large surface area to volume ratio - must make reference to 'walls' ACCEPT single layer of squamous epithelial cells ACCEPT lots of capillaries / many capillaries / surrounded by capillaries / covered by capillaries	(5)

- The structure and properties of the cell membrane control which molecules can move into or out of the cell.

(a) The phospholipid bilayer plays an important role in this control of movement of molecules.

Explain why the phospholipid molecules form a bilayer.

Question Number	Answer	Additional Guidance	Mark
3(a)	{phosphate group / heads} are hydrophilic ; - Idea that heads can be attracted to water ; {fatty acids / tails} are hydrophobic ; - Idea that tails orientate themselves away from water / eq ; - Idea of aqueous environment on both sides of the membrane ;	ACCEPT marks for annotated diagram, phonetic spelling OK IGNORE "water loving / hating" - CCEPT polar - t just facing water - CCEPT non polar - CCEPT repel water, face away from water, away from polar environment - CCEPT polar environment	(3)

- Describe the structure of a cell membrane.

Question Number	Answer	Additional Guidance	Mark
4(a)	- reference to phospholipid bilayer ; - correct orientation and structure of the phospholipids in the bilayer ; - explanation of why the phospholipids are orientated the way they are e.g. heads attracted to water OR tails repelled by water ; - proteins in the membrane (described / shown) ; - idea of two different locations of proteins e.g. extrinsic, intrinsic, transmembrane ; - glycoproteins / glycolipids (described / shown) ; - idea of cholesterol within the membrane (described / shown) ;	Read what is written on the lines first Accept points made on a clearly labelled diagram If diagram and description contradict then Mp not awarded 2. CCEPT heads on outside and each with two tails if drawn 2. N if gap between phospholipids is too large e.g. as large as a phospholipid in the diagram 3. CCEPT ref to heads being hydrophilic OR tails hydrophobic OR explained in terms of polarity 5. If only one protein located then still get Mp4	(5)

- Use your knowledge of the structure and properties of cell membranes to explain the permeability of non-polar molecules

Question Number	Answer	Mark
3(b)(iv)	- reference to phospholipid bilayer ; - reference to hydrophobic nature (of bilayer / tails) ; - idea that {non-polar molecules / molecules that have high solubility in oil compared with water} will pass through the membrane more readily OR idea that {polar molecules / molecules with low solubility in oil relative to water} will pass through less readily ; - idea that permeability linked to readiness to dissolve ; - reference to {fluidity / movement} of phospholipids ;	max (3)

- Explain how the properties of molecule A (Phospholipid) contribute to the structure of the cell membrane.

Question Number	Answer	Mark
4(a)(ii)	- reference to {hydrophilic / polar / charged} part ; - reference to {hydrophobic / non polar / uncharged} part ; - reference to orientation of molecule in relation to water; - idea that aqueous environment is {on two sides / cytoplasm and {environment / tissue fluid / eq}} ;	max (3)

- Some proteins in the cell membrane are involved in active transport and facilitated diffusion. Describe the role of proteins in these cell transport mechanisms.

Question Number	Answer	Mark
4(b)	Active transport: - idea that molecule {binds / fits into} {protein / carriers} ; - idea that {protein / carrier} changes shape ; (molecules move) against a concentration gradient / eq ; - reference to use of {ATP / energy} ; [Submax 2 marks] Facilitated diffusion: - reference to proteins as {channels / gates / pores / carriers} ; - idea that {channels can open or close / carriers change shape} ; - for {large / polar / charged} molecules (to pass through membrane) ; (molecules move) down a concentration gradient / eq ; [Submax 2 marks]	max (3)

Question Number	Answer	Mark
4(c)(ii)	- idea that {phospholipids / molecule A} allow {fluidity / movement/ eq} ; - idea that {fluidity / movement / eq} allow membranes to fuse; - idea that {fluidity / movement / eq} allows protein to {move / intermingle / eq} ;	max (2)

- Explain how the circulatory system in mammals enables efficient gas exchange.

Question Number	Answer	Additional Guidance	Mark
4(b)	- idea that mass flow generated by heart ; - idea that moving blood helps to maintain a concentration gradient ; - idea that a { steep / eq } concentration gradient gives a { fast / eq } rate of diffusion ; - idea of { network / lots / eq } of capillaries ; - large surface area of capillaries / eq ; - idea that large surface area increases the rate of diffusion ; - idea that capillaries have very thin walls ; - idea that diffusion is fastest over small distances ; - idea that no {organs / cells / tissues } are far away from blood / capillaries cover alveoli / eq ; - idea that efficiency is related to double circulation ;	- IGNORE thin capillaries - ACCEPT references to Fick's law	(5)

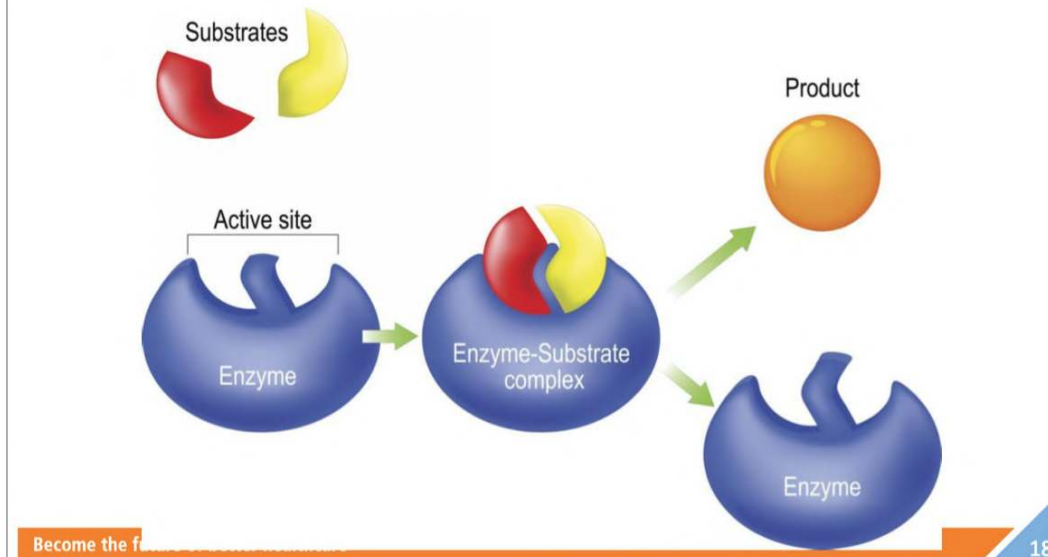
Enzymes

▼ What are enzymes?

- Enzymes are globular proteins that act as biological catalysts by speeding up chemical reactions and remain unchanged at the end of the reaction.
- The active site is at the **TERTIARY** structure of the enzyme.
- Made inside all living organisms - extracellular: functions outside of the cell & intracellular - functions within the cell.

▼ Describe the mode of enzyme activity

Enzyme Structure and function



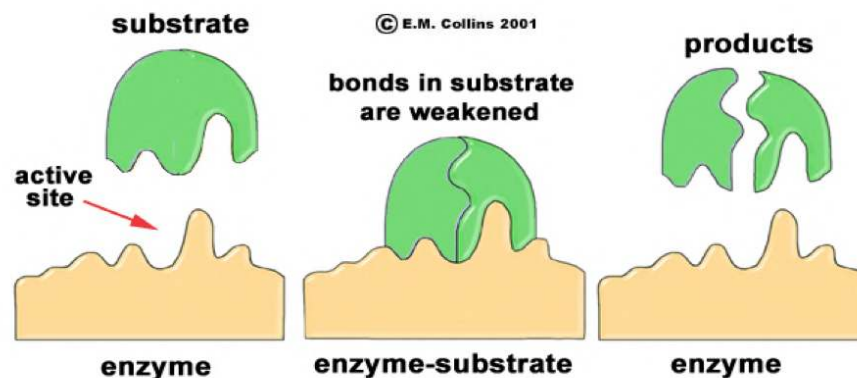
▼ What are the properties of an enzyme?

1. Globular Protein
2. Lowers activation energy
3. Regulated by pH changes
4. Denatures
5. Reaction is reversible
6. Highly specific
7. Needed in low amounts
8. Need cofactors & coenzymes

▼ What is the active site of an enzyme?

- Region where substrate molecules bind and undergo a chemical reaction
- Has a specific shape so only specific substrates can fit
- Has high affinity and is complementary in shape to substrates
- Weak and temporary bonds are formed between the substrate & enzyme to form an enzyme - substrate complex

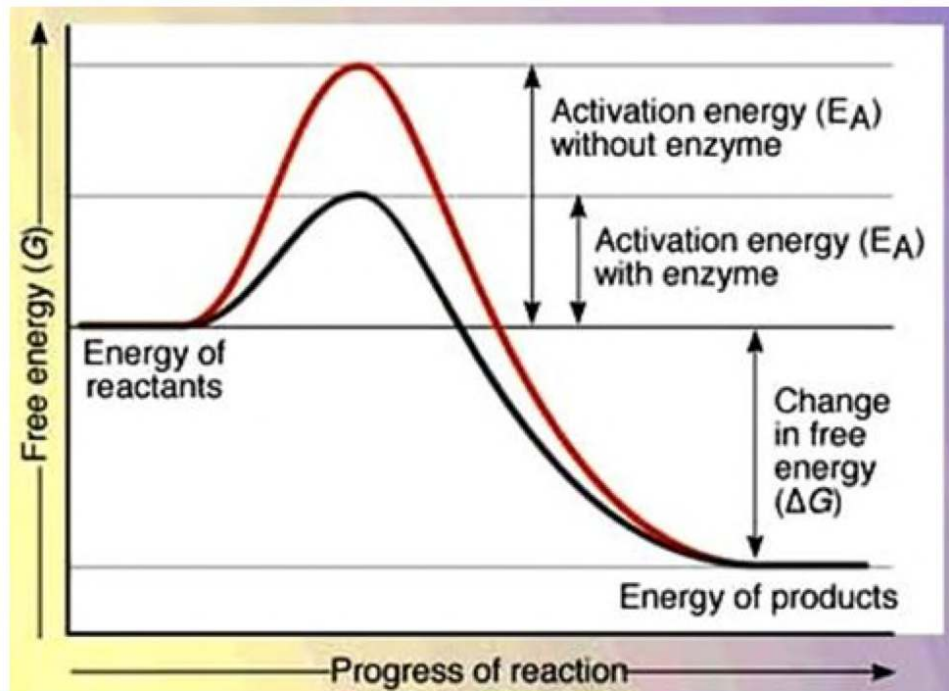
How enzyme works with substrate?



- One to one reaction
- Interaction between enzyme-substrate break forming products, then products leave active site
- enzyme unchanged

▼ What is activation energy? How does activation energy changed by an enzyme affect a reaction?

- Enzyme needed for substrate to react
- It is needed to break or make bonds
- Can be provided in heat from that substrate gets from surroundings
- Enzymes speed up reactions by lowering this



▼ Describe the "Lock and Key" Method

The active site is specific and complementary to a specific substrate molecule. Thus only one type of substrate can combine with the active site to form an enzyme-substrate complex. The substrate acts as a key while the active site is the keyhole and the enzyme is the lock for the substrate to fit in.

The substrate fits into the active site, the bonds are broken/formed depending on whether it is breaker or builder enzymes and the products leave the active site when done.

▼ Describe the "Induced Fit" Method of enzyme activity

The induced fit model is different as it states that the active site is more flexible. When the substrate approaches the active site, the active site's shape & charge manipulate the substrate into fitting into the site - thereby 'inducing' a fit.

The active site changes its shape while being attached to the substrate to form a complex and releasing the products when the reaction is completed.

The active site & enzyme goes back to their original states when the reaction is over.

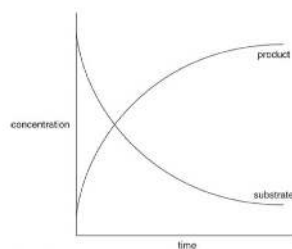
▼ How do we measure enzyme activity or rate?

1. Disappearance of substrate over time
2. Formation of product over time



How to measure enzyme activity??

Eg: catalase activity



Graph shows:

- a) Disappearance of substrate over time
- b) Appearance of product over time

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Factors affecting Enzyme Activity:

▼ What are the 4 factors affecting enzyme activity?

- temperature
- pH
- substrate concentration
- enzyme concentration

▼ What happens if pH is not optimum?

If it is not optimum, it disrupts the ionic bonds. Tertiary structure is disrupted and enzyme loses their shape and function and eventually becomes denatured.

▼ What happens if a) temp is too low? b) temp is too high?

At low temperatures, there are less collisions between substrate molecules and active sites so there are less complexes formed.

Enzymes can be preserved at low temperatures since activity becomes dormant then.

At increasing temperatures (20C) - the kinetic energy increases, so more collisions happens and thus more complexes are formed. The rate of reaction doubles for every 10 degree increase.

Enzymes work best at 37C to 40C- rate of product formation is the highest.

After 40C, there is high kinetic energy that breaks hydrogen bonds in the active site so the enzyme gets denatured as it loses its shape.



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